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B.Tech. Degree VI Semester Examination April 2018

ME 15-1602 MACHINE DESIGN I (2015 Scheme)

Time : 3 Hours

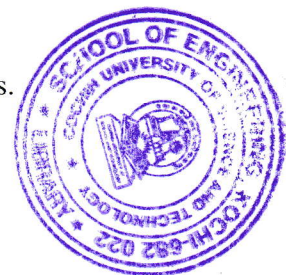
Maximum Marks : 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) What are the different steps involved in the design of a machine element?
- (b) What is the difference between failure due to static load and fatigue failure?
- (c) State and explain the term 'Resilience'.
- (d) Explain the thread nomenclature with neat sketch.
- (e) Explain the condition of self-locking of power screw.
- (f) Enumerate the different types of riveted joints and rivets.
- (g) Discuss the materials and practical applications for the various types of springs.
- (h) What is the objective of nipping of leaf spring?
- (i) What are the assumptions made in the design of welded joint?
- (j) Explain the term 'critical speed of shaft'.



PART B

(4 × 10 = 40)

- II. A hot rolled steel shaft is subjected to a torsional moment that varies from 330 Nm clockwise to 110 Nm counter clockwise and an applied bending moment at a critical section varies from 440 Nm to 220 Nm. Determine the required shaft diameter. The material has an ultimate strength of 550 MN/m^2 and a yield strength of 410 MN/m^2 . Take the endurance limit as half the ultimate strength, F.O.S = 2, size factor = 0.85, surface finish factor = 0.62, load correction factor = 1.0 in bending and 0.55 in torsion.

OR

- III. A steel shaft having cylindrical cross section of yield strength 800 MPa is subjected to static loads consisting of bending moment 15 kNm and torsional moment 45 kNm. Determine the diameter of the shaft using (i) Maximum shear stress theory (ii) Maximum strain energy theory of failure. Take $E = 210 \text{ GPa}$, Poisson's ratio = 0.25. Assume factor of safety of 2.

- IV. Two 35 mm diameter shafts are connected by a flanged coupling. The flanges are fitted with 6 bolts on 125 mm bolt circle. The shafts transmit a torque of 800 Nm at 350 rpm. For the safe stresses mentioned below, calculate (i) diameter of bolts (ii) thickness of flanges (iii) key dimensions (iv) hub length (v) power transmitted.

Safe shear stress for shaft material	=	63 MPa.
Safe stress for bolt material	=	56 MPa.
Safe stress for cast iron material	=	10 MPa.
Safe stress for key material	=	46 MPa.

OR**(P.T.O.)**

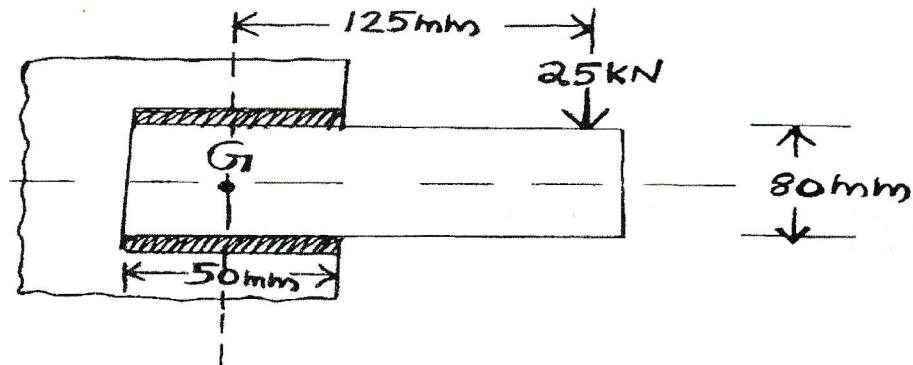
- V. Design a gib and cotter joint to connect two mild steel rods for a pull of 30 kN . The maximum permissible stresses are 55 MPa in tension; 40 MPa in shear and 70 MPa in crushing. Draw a neat sketch of the joint designed.

- VI. Design a triple riveted longitudinal double strap butt joint with unequal straps for a boiler. The inside diameter of the longest course of the drum is 1.3 m . The joint is to be designed for a steam pressure of 2.4 N/mm^2 . The working stresses to be used are 77 MPa in tension, 62 MPa in shear and 120 MPa in crushing. Assume the efficiency of joint as 81% .

OR

- VII. A helical compression spring made of oil tempered carbon steel, is subjected to a load which varies from 600 N to 1600 N . The spring index is 6 and the design factor of safety is 1.43 . If the yield shear stress is 700 MPa and the endurance stress is 350 MPa , find the size of the spring wire and mean diameter of the spring coil.

- VIII. A bracket carrying a load of 25 kN is to be welded as shown in figure. Find the size of weld required if the allowable shear stress is not to exceed 80 MPa .



OR

- IX. A shaft made of mild steel is required to transmit 100 kW at 300 rpm . The supported length of the shaft is 3 m . It carries two pulleys each weighing 1500 N supported at a distance of 1 m from the end respectively. Assuming the safe value of stress, determine the diameter of the shaft.
